

SLIDE 1: TITLE & PRESENTATION DETAILS

B.Sc. Graduation Project Defense Presentation

- **Project Title:** DESIGN AND IMPLEMENTATION OF AN INTELLIGENT DOCUMENT ANALYSIS AND LEARNING ASSISTANT SYSTEM (AuraQA)
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- **Supervision:** Dr. David Oyewola (Supervisor) | Dr. (Mrs.) Asabe (HOD)
- **Degree & Session:** Bachelor of Science (B.Sc.) Computer Science Defense Presentation (July, 2026)

SLIDE 2: INTRODUCTION

Abstract, Problem Statement, Aim & Objectives

- **Short Abstract:** AuraQA is a fully offline RAG system that eliminates cloud data privacy risks and API costs while optimizing CPU prefill latency.
- **Problem Statement:** Manual document reading is inefficient, keyword search lacks context, and cloud LLMs expose sensitive data to privacy risks.
- **Aim of the System:** Develop a high-speed, secure, offline Intelligent Document Analysis and Learning Assistant System using local RAG.
- **Objectives:** 1. Support PDF, DOCX, TXT, CSV; 2. Local QA via Ollama (Qwen/Gemma); 3. Offline Summaries, Quizzes & Flashcards; 4. Out-of-Scope Shield & Hausa Support.

SLIDE 3: LITERATURE REVIEW

Theoretical Framework, RAG Paradigm & Research Gap

- **Theoretical Framework:** Retrieval-Augmented Generation (RAG) combines dense vector search with local generative models to eliminate AI hallucinations.
- **RAG Paradigm:** paraphrase-multilingual-MiniLM-L12-v2 generates 384-dim embeddings; FAISS FlatIP index enables instant cosine similarity retrieval.
- **Multilingual Foundation:** Native cross-lingual vector alignment allows querying English documents in Hausa, Arabic, French, Spanish, and German.
- **Research Gap:** Bridging the gap between cloud-dependent LLMs and zero-cost, resource-constrained CPU offline hardware.

SLIDE 4: METHODOLOGIES

SDLC Rationale, Ingestion Pipeline & Tech Stack

- **Why Prototyping SDLC?** 1. Iterative Chunking (benchmarked 1000-char chunks + 150-char overlap); 2. UI Refinement (collapsible sidebar & study tools); 3. Latency Optimization (0.00s router).
- **Ingestion & Extraction:** Multi-format text extraction (PDF, DOCX, TXT, CSV) mapped to exact page/row metadata for citation tracking.
- **Deterministic Classifier:** Rule-based query router bypasses slow LLM classification calls, executing intent routing in 0.00 seconds.
- **Tools & Tech Stack:** Next.js 16 + Tailwind CSS (Frontend), FastAPI + SQLite (Backend), FAISS (Vector Store), Ollama Qwen 2.5/Gemma 2 (Local LLM).

SLIDE 5: SYSTEM DESIGN, IMPLEMENTATION & RESULTS

Architecture, Key Features & Benchmark Evaluation

- **System Architecture:** User → Next.js SPA → FastAPI Server → Sentence Transformers → FAISS Index → Ollama LLM → Grounded Response.
- **Key Feature 1 (Grounded QA & Chat):** Interactive chat with Gemini-style sidebar, streaming responses, and out-of-scope grounding shield (0.40 threshold).
- **Key Features 2, 3 & 4 (Study Mode & Hausa):** Automated Summary extraction, Multiple-Choice Quiz generator, Study Flashcards, and Hausa (`ha`) cross-lingual QA.
- **Benchmark Results:** Overall RAG CPU latency reduced by 48.57% (71.69s to 36.87s); router latency 0.00s; 100% out-of-scope hallucination block.

SLIDE 6: SUMMARY, CONCLUSION & RECOMMENDATIONS

Achievements, Institutional Significance & Future Work

- **Summary of Achievements:** Built a fully functional, zero-cost, privacy-preserving offline learning assistant operating locally on standard laptop CPUs.
- **Conclusion:** Proves that data sovereignty and high-performance AI document analysis are achievable without cloud API dependency.
- **Recommendations:** Deploy AuraQA across university libraries, research departments, and schools with limited internet connectivity.
- **Future Extensions:** Incorporate OCR for scanned documents, mobile runtime (ONNX), and local Whisper voice assistant interaction.

SLIDE 7: REFERENCES & DEFENSE CONCLUSION

Short Academic Literature & Panel Q&A;

- **Lewis et al. (2020):** Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks. NeurIPS.
- **Johnson et al. (2019):** Billion-scale similarity search with GPUs. IEEE Transactions on Big Data (FAISS Framework).
- **Reimers & Gurevych (2019):** Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks. EMNLP.
- **Vaswani et al. (2017):** Attention Is All You Need. NeurIPS (Transformer Architecture).
- **Acknowledgement & Q&A;:** Heartfelt thanks to FUK, Supervisor Dr. David Oyewola, HOD Dr. (Mrs.) Asabe. Open for panel questions.